COMP3000HK Computing Project

2025/2026

Finalized Project Title

A Machine Learning Approach for Identifying Psychological Triggers in Suspicious Emails

Links

GitHub repository: https://github.com/hkuspace-pu/project-wongoining.git

Live prototype: https://phishing-trigger-analyser.streamlit.app/

Demonstration video: https://youtu.be/EW3MCXhEWuY

Finalised Project Description

This project develops a machine learning-based system for identifying psychological triggers in suspicious emails. Rather than functioning as a first-layer phishing detector that only determines whether an email is safe or unsafe, the system is designed as a second-stage phishing awareness support tool. It analyses suspicious email text and predicts seven psychological trigger categories: Urgency, Scarcity, Authority, Fear, Social Proof, Reciprocity, and Liking.

A manually annotated dataset of 977 phishing and scam emails was used to train and evaluate multiple approaches, including rule-based keyword matching, traditional TF-IDF machine learning models, DistilBERT, and RoBERTa. The final selected model was RoBERTa, which achieved the strongest five-fold cross-validated Macro F1 performance. A Streamlit prototype was developed to present trigger confidence scores, explanations, and protective guidance, helping users understand how suspicious emails may attempt to influence their judgement.

Updated Keywords

Phishing Awareness; Psychological Triggers; Social Engineering; Machine Learning; Multi-label Classification; RoBERTa; NLP; Email Security; Suspicious Emails; Cybersecurity

This document is provided as supporting evidence for the Project & Description section.